

Numerical Methods

By

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Newton-Raphson Method

The Newton-Raphson method is the best-known method of finding roots of a function $f(x)$. The method is simple and fast.

Basic Idea:

The method uses:

- ☐ The function $f(x)$
- ☐ Its derivative $f'(x)$

It starts with an initial guess near the root. Hence, the Newton-Raphson method is usable only in problems where $f'(x)$ can be readily computed.

Newton-Raphson method is also called Newton's method

Algorithm for Newton-Raphson Method

Step 1: Choose an initial approximation x_0 close to the actual root.

Step 2: Compute the next approximation using:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, n = 0, 1, 2, \dots$$

Step 3: Repeat **Step 2** until one of the stopping criteria is satisfied:

$$|x_{n+1} - x_n| < \epsilon$$

OR:

$$|f(x_{n+1})| < \epsilon$$

Example :

Use Newton-Raphson method to find the real root near 2 of the equation

$$x^4 - 11x + 8 = 0$$

accurate to five decimal places.

Solution :

Here,

$$f(x) = x^4 - 11x + 8$$

$$f'(x) = 4x^3 - 11$$

The initial guess of the solution $x_0 = 2$

$$f(x_0) = f(2) = 2^4 - 11(2) + 8 = 2$$

$$f'(x_0) = f'(2) = 4(2)^3 - 11 = 21$$

Therefore,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 - \frac{2}{21} = 1.90476$$

$$|x_1 - x_0| = |1.90476 - 2| = 0.0952381$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1.90476 - \frac{(1.90476)^4 - 11(1.90476) + 8}{4(1.90476)^3 - 11} = 1.89209$$

$$|x_2 - x_1| = |1.89209 - 1.90476| = 0.0126697$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.89209 - \frac{(1.89209)^4 - 11(1.89209) + 8}{4(1.89209)^3 - 11} = 1.89188$$

$$|x_3 - x_2| = |1.89188 - 1.89209| = 0.000216$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 1.89188 - \frac{(1.89188)^4 - 11(1.89188) + 8}{4(1.89188)^3 - 11} = 1.89188$$

$$f(x_4) = f(1.89188) = (1.89188)^4 - 11(1.89188) + 8 = 0.000064$$

Hence the root of the equation is 1.89188.

Example :

Use Newton-Raphson method to find the real root near 1 of the equation

$$e^x - 3x^2 = 0$$

accurate to five decimal places.

Solution :

Here,

$$f(x) = e^x - 3x^2$$

$$f'(x) = e^x - 6x$$

The initial guess of the solution $x_0 = 1$

$$f(x_0) = f(1) = e^1 - 3(1)^2 = -0.281718$$

$$f'(x_0) = f'(1) = e^1 - 6(1) = -3.28172$$

Therefore,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1 - \frac{0.281718}{3.28172} = 0.91416$$

$$|x_1 - x_0| = |0.91416 - 1| = 0.085845$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.91416 - \frac{e^{0.91416} - 3(0.91416)^2}{e^{0.91416} - 6(0.91416)} = 0.91002$$

$$|x_2 - x_1| = |0.91002 - 0.91416| = 0.00414$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.91002 - \frac{e^{0.91002} - 3(0.91002)^2}{e^{0.91002} - 6(0.91002)} = 0.910008$$

$$|x_3 - x_2| = |0.910008 - 0.91002| = 0.00001$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 0.910008 - \frac{e^{0.910008} - 3(0.910008)^2}{e^{0.910008} - 6(0.910008)} = 0.910008$$

$$f(x_4) = f(0.910008) = e^{0.910008} - 3(0.910008)^2 = -1.791 \times 10^{-10}$$

Hence the root of the equation is 0.91001.

Example :

Evaluate $\sqrt{29}$ to five decimal places by [Newton-Raphson iterative method](#).

Solution :

Let, $x = \sqrt{29}$ then $x^2 - 29 = 0$

$$f(x) = x^2 - 29$$

$$f'(x) = 2x$$

Now $f(5) = 25 - 29 = -4 < 0$ and $f(6) = 36 - 29 = 7 > 0$.
Hence, a root of $f(x) = 0$ lies between 5 and 6.

The initial guess of the solution $x_0 = 5$

$$f(x_0) = f(5) = 5^2 - 29 = -4$$

$$f'(x_0) = f'(5) = 2(5) = 10$$

Therefore,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 5 - \frac{-4}{10} = 5.4$$

$$|x_1 - x_0| = |5.4 - 5| = 0.4$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 5.4 - \frac{(5.4)^2 - 29}{2(5.4)} = 5.38519$$

$$|x_2 - x_1| = |5.38516 - 5.4| = 0.014815$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 5.38519 - \frac{(5.38519)^2 - 29}{2(5.38519)} = 5.38516$$

$$|x_3 - x_2| = |5.38516 - 5.38519| = 0.00002$$

$$f(x_3) = f(5.38516) = (5.38516)^2 - 29 = 4.2 \times 10^{-10}$$

Hence the root of the equation is 5.38516.

Secant Method

The Secant Method is an iterative technique used to find a root of a nonlinear equation

$$f(x) = 0$$

It is similar to the Newton–Raphson method, but it does not require the derivative.

the derivatives of the function being approximated by finite differences instead of being calculated analytically.

In particular, the derivative $f'(x)$ is approximated by the backward difference

$$f'(x) = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$$

where x_n and x_{n-1} are two approximations to the root but does not require the condition $f(x_i) \cdot f(x_{i-1}) < 0$.

Now, from the Newton-Raphson method, we have

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Algorithm for Secant Method

Step 1: Choose two initial approximations x_0 and x_1 close to the root

Step 2: Compute the next approximation using:

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}, n = 1, 2, 3, \dots$$

Step 3: Repeat **Step 2** until one of the stopping criteria is satisfied:

$$|x_{n+1} - x_n| < \epsilon$$

OR:

$$|f(x_{n+1})| < \epsilon$$

Example :

Use Secant method to find the root of the equation

$$x^3 - 8x - 5 = 0$$

accurate to five decimal places Let the initial approximations be $x_0 = 3$, and $x_1 = 3.5$

Solution :

$$f(x) = x^3 - 8x - 5$$

$$f(x_0) = f(3) = 3^3 - 8(3) - 5 = -2$$

$$f(x_1) = f(3.5) = (3.5)^3 - 8(3.5) - 5 = 9.875$$

Therefore,

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

$$x_2 = x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = 3.5 - \frac{(9.875)(3.5 - 3)}{(9.875) - (-2)} = 3.08421$$
$$|x_2 - x_1| = |3.08421 - 3.5| = 0.41579$$

$$f(x_2) = f(3.08421) = (3.08421)^3 - 8(3.08421) - 5 = -0.33558$$

$$x_3 = x_2 - \frac{f(x_2)(x_2 - x_1)}{f(x_2) - f(x_1)}$$
$$x_3 = (3.08421) - \frac{(0.33558)(3.08421 - 3)}{(0.33558) - (9.875)} = 3.09788$$
$$|x_3 - x_2| = |3.09788 - 3.08421| = 0.01367$$

$$f(x_3) = f(3.09788) = (3.09788)^3 - 8(3.09788) - 5 = -0.05320$$

$$x_4 = x_3 - \frac{f(x_3)(x_3 - x_2)}{f(x_3) - f(x_2)}$$

$$x_4 = 3.09788 - \frac{(-0.05320)(3.09788 - 3.08421)}{(-0.05320) - (-0.33558)} = 3.10045$$

$$|x_4 - x_3| = |3.10045 - 3.09788| = 0.00257$$

$$f(x_4) = f(3.10045) = (3.10045)^3 - 8(3.10045) - 5 = 0.00039$$

$$x_5 = x_4 - \frac{f(x_4)(x_4 - x_3)}{f(x_4) - f(x_3)}$$

$$x_5 = (3.10045) - \frac{(0.00039)(3.10045 - 3.09788)}{(0.00039) - (-0.05320)} = 3.10043$$

$$|x_5 - x_4| = |3.10045 - 3.10043| = 0.00002 = 2 \times 10^{-5}$$

Hence, a root is 3.1004 correct up to five significant figures

Example :

Determine a root of the equation $e^x - 3x = 0$ using [the secant method](#).
The initial approximations $x_0 = 0$, and $x_1 = 1$

Solution :

$$f(x) = e^x - 3x$$

$$f(x_0) = f(0) = e^0 - 3(0) = 1$$

$$f(x_1) = f(1) = e^1 - 3(1) = -0.2817$$

Therefore,

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

$$x_2 = x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = 1 - \frac{(-0.2817)(1 - 0)}{(-0.2817) - (1)} = 0.7802$$
$$|x_2 - x_1| = |0.7802 - 1| = 0.2198$$

$$f(x_2) = f(0.7803) = e^{(0.7803)} - 3(0.7803) = -0.1587$$

$$x_3 = x_2 - \frac{f(x_2)(x_2 - x_1)}{f(x_2) - f(x_1)}$$
$$x_3 = 0.7803 - \frac{(-0.1580)(0.7803 - 1)}{-0.1580 - (-0.2817)} = 0.4966$$

$$|x_3 - x_2| = |0.4966 - 0.7802| = 0.2836$$

$$f(x_3) = f(0.4966) = e^{(0.4966)} - 3(0.4966) = 0.15333$$

$$x_4 = x_3 - \frac{f(x_3)(x_3 - x_2)}{f(x_3) - f(x_2)}$$

$$x_4 = 0.4966 - \frac{(0.15333)(0.4966 - 0.7802)}{(0.15333) - (-0.1587)} = 0.63596$$
$$|x_4 - x_3| = |0.63596 - 0.4966| = 0.1394$$

$$f(x_4) = f(0.63596) = e^{(0.63596)} - 3(0.63596) = -0.01905$$

$$x_5 = x_4 - \frac{f(x_4)(x_4 - x_3)}{f(x_4) - f(x_3)}$$
$$x_5 = (0.63596) - \frac{(-0.01905)(0.63596 - 0.4966)}{(-0.01905) - (0.15333)} = 0.62056$$

$$|x_5 - x_4| = |0.62056 - 0.63596| = 0.0154$$

$$f(x_5) = f(0.62056) = e^{(0.62056)} - 3(0.62056) = -0.0017$$

$$x_6 = x_5 - \frac{f(x_5)(x_5 - x_4)}{f(x_5) - f(x_4)}$$

$$x_6 = 0.62056 - \frac{(-0.0017)(0.62056 - 0.63596)}{(-0.0017) - (-0.01905)} = 0.61905$$

$$|x_6 - x_5| = |0.61905 - 0.62056| = 0.0015$$

$$f(x_6) = f(0.61905) = e^{(0.61905)} - 3(0.61905) = 0.000013$$

Hence, a root is **0.61905** correct up to five significant figures

**THANK YOU
FOR YOUR
ATTENTION**

